

COMICCRAFT – REQUIREMENT ANALYSIS PHASE

AI Comic Story Creator using Gemini Models

1. Introduction

The Requirement Analysis Phase identifies the functional and technical requirements needed to develop ComicCraft. The application is intended to accept user-defined comic preferences, generate a structured story, create corresponding illustrations, display the comic in a web interface, and provide a downloadable PDF.

2. Project Objectives

- Generate personalized comic stories from user-provided prompts.
- Generate a structured five-panel comic outline.
- Generate narration and character dialogues for the comic.
- Generate comic-style illustrations for individual panels.
- Display the generated comic panel-by-panel in a web interface.
- Export the completed comic as a PDF.

3. Functional Requirements

FR-01: User Input

The system shall provide an input form for story prompt, main character name, setting, story tone, and art style.

FR-02: Story Outline Generation

The system shall generate a structured five-panel comic outline from the user's story prompt.

FR-03: Story Narration and Dialogue

The system shall expand the panel outline into comic narration and character dialogue.

FR-04: Image Generation

The system shall generate a comic-style illustration for each panel using the corresponding image prompt.

FR-05: Comic Layout

The system shall associate generated images with their corresponding panel stories and organize them into a structured layout.

FR-06: Comic Preview

The system shall display the generated comic sequentially in the web interface.

FR-07: PDF Export

The system shall compile panel images and narrative content into a multi-page PDF.

FR-08: Export Confirmation

The system shall display an export-success page after the comic PDF is generated/downloaded.

FR-09: JSON API

The system shall support JSON-based comic generation through the `/generate-comic/json` route.

FR-10: Image Testing

The system shall provide a `/test-image` route for testing image generation independently.

4. User Input Requirements

Story Prompt: Main idea or concept of the comic.

Main Character Name: Name of the hero/main character.

Setting: Location or environment such as forest, school, city, or space.

Story Tone: Mood such as light-hearted, dramatic, poetic, or funny.

Art Style: Visual preference such as anime, pixel art, comic book, or realistic.

5. AI and Processing Requirements

Gemini Flash: Used for fast, structured generation of the five-panel comic outline.

Gemini Pro: Used for detailed comic narration and character dialogues.

Stable Diffusion: Used for generating comic-style illustrations from image prompts.

6. Backend Requirements

- FastAPI shall manage server-side routing and request processing.
- Form data shall be captured through FastAPI form parameters.
- JSON requests shall be validated using a Pydantic request model.
- The backend shall coordinate outline generation, story generation, image generation, layout building, and PDF export.
- The backend shall handle unexpected processing/API errors using appropriate exception handling.

7. Frontend Requirements

- The frontend shall use HTML, CSS, and Jinja2 templates.

- index.html shall collect the user's comic preferences.
- comic_preview.html shall display generated comic panels.
- export_success.html shall provide confirmation after export.
- The interface shall present panel title, illustration, scene description, captions, and narration.

8. PDF Export Requirements

- The system shall use the generated panel images and narrative content.
- Each panel's image and narration shall be organized into the PDF.
- The PDF shall be generated using the FPDF library.
- The generated PDF shall be saved in the static/exports folder according to the project documentation.

9. Non-Functional Requirements

Usability: The interface should provide a simple flow from entering a story idea to viewing and exporting the comic.

Maintainability: The application should separate frontend, backend, AI integration, layout, and export responsibilities into appropriate modules.

Performance: The selected AI workflow should support efficient structured story generation and image generation.

Reliability: The backend should handle API and processing failures gracefully.

Extensibility: The modular architecture should allow future additions such as user profiles, comic libraries, and multi-page story arcs.

10. Software and Environment Requirements

- Python with pip.
- FastAPI and Uvicorn.
- Jinja2 and python-multipart.
- Google Generative AI integration.
- Hugging Face Diffusers and Transformers.
- PyTorch-related image-generation dependencies as required by the selected Diffusers workflow.
- FPDF for PDF generation.
- Pillow, Accelerate, and other project dependencies.
- Git for version control.
- A virtual environment for isolated project dependencies.

11. API and Route Requirements

/: Loads the ComicCraft homepage.

/generate: Processes form input and runs the complete comic-generation workflow.

/generate-comic/json: Accepts JSON input and returns comic layout data and PDF path.

/test-image: Tests image generation using a direct prompt.

/export-success: Displays confirmation after comic export.

12. Input-Process-Output Analysis

Input:

- Story prompt
- Character name
- Setting
- Story tone
- Art style

Process:

- Generate five-panel outline.
- Generate narration and dialogue.
- Generate panel illustrations.
- Build comic layout.
- Generate PDF.

Output:

- Panel-by-panel comic preview.
- Generated comic images.
- Narration and dialogue.
- Downloadable comic PDF.

13. Requirement Traceability to Project Workflow

- Model selection and architecture requirements support Milestone 1.
- Core AI functionality and FastAPI processing support Milestone 2.
- Route requirements support Milestone 3.
- Frontend and Jinja2 requirements support Milestone 4.
- Local deployment and testing requirements support Milestone 5.

14. Expected Result of Requirement Analysis

The requirement analysis establishes the necessary user inputs, AI processing stages, backend routes, frontend pages, image generation workflow, comic layout, PDF export, and deployment requirements needed to implement the ComicCraft application.

15. Reference

Prepared from the provided project documentation: "ComicCraft - AI Comic Story Creator using Gemini Models".